

Igor Garcia

Software Engineer | EU Citizen | +55(75)999823805 | igonildo7@gmail.com

LINKS

[Personal website](#) , [Github](#) , [LinkedIn](#)

Software Engineer with 4+ years of experience building high-performance internal tooling, automated workflows, and network-aware applications. Strong background in C++ and C#, with a proven track record of optimizing complex systems and reducing manual operations for engineering teams. Deeply passionate about systems engineering, backend architecture, and cloud infrastructure. Currently leveraging AWS services to build scalable, automated solutions, with hands-on experience in containerized deployments (Docker) and CI/CD automation. EU Citizen open to relocation (Ireland/UK/EU) or remote work.

SKILLS

Languages: C++, Python, C#, TypeScript, SQL
Systems & Networking: HTTP, TCP/IP fundamentals, RESTful APIs, Network Replication, Systems Optimization, Concurrent Programming
Cloud & Backend: AWS (Lambda, API Gateway, DynamoDB, SQS, CloudWatch), ASP.NET Core, Serverless Architecture, Microservices
DevOps & Tools: Docker, Docker Compose, CI/CD (GitHub Actions – build, test, deploy pipelines), Git, Linux/OS fundamentals, Automated Tooling
Additional (background): Unreal Engine 5 (C++/Blueprint), Unity (C#), multiplayer networking/replication, profiling/optimization

WORK EXPERIENCE

Virtual Reality Researcher, Ford Motor Company

Aug 2021 — Dec 2022

- Spearheaded R&D for automotive VR applications, establishing foundational workflows that were later adopted as the **company standard for virtual prototyping**.
- Collaborated with Design and Engineering teams to translate physical requirements into virtual specifications, resulting in user assessment accuracy.
- Developed initial prototypes for virtual user experience testing, validating key interaction concepts for future production.

Software engineer/C++ developer, Ford Motor Company

Jan 2023 — Present

- Engineered high-performance internal software tooling and automated validation pipelines using **C++**, replacing physical prototypes and contributing to **material cost reduction**.
- Collaborated closely with internal customers (Design and Engineering teams) to define and build the tools they need, reducing design iteration cycles.
- Optimized complex visualizations and system performance across multiple hardware targets, sustaining stable frame rates while processing high-density assets (5M+ data points/polygons).
- Delivered robust software features with a strong focus on performance profiling, memory management, stability, and maintainability.

Unreal Engine 5 Developer, Blue Gravity Studios

Jan 2024 — Sep 2024

- Engineered core network replication logic (TCP/UDP concepts) for multiplayer architecture using **C++**, optimizing bandwidth usage to support highly concurrent online sessions.
- Developed core software systems, improving code modularity and reducing bug-fixing overhead by **20%** for the engineering team.
- Implemented scalable input systems and ensuring 100% compatibility across multiple platforms.
- Collaborated with multidisciplinary teams to optimize memory usage and processing calls, enhancing overall performance on lower-end hardware.

Software engineer, Cafundo Creative Studio

Jun 2025 — Present

- Architected an AI-powered interactive totem in Unity (C#) for high-traffic public environments, serving **over 500 daily interactions** with zero downtime.
- Seamlessly integrated Azure Cognitive Services and OpenAI APIs, reducing voice-to-response latency by **30%** to create a natural conversational flow.
- Designed intuitive UI/UX systems focused on accessibility and seamless real-time user interaction.

PROJECTS (All project pages accessible from [Portfolio](#))

Resume Matcher — AI Job Search Platform | C# / .NET 9 / React / TypeScript / PostgreSQL 2026

A full-stack platform that aggregates remote jobs from 7+ boards and uses Google Gemini AI to score resume-to-job compatibility.

- Designed a clean-architecture .NET 9 API with Strategy pattern to query 7 job providers concurrently with deduplication fallback.
- Integrated Google Gemini AI to deliver match scores, keyword gaps, and improvement suggestions.

- Implemented secure auth (ASP.NET Identity + JWT in HttpOnly cookies) with user profiles and persistent analysis history.
- Deployed via Docker on Render (backend) and Vercel (frontend) with PostgreSQL persistence.

PSN Price Tracker API & Bot — C# / .NET 9 / Docker 2026

A backend system built to continuously monitor PlayStation Store prices and dispatch real-time alerts via Telegram.

- **Architecture & Concurrency:** Designed an asynchronous backend utilizing IHostedService to manage concurrent background workers for continuous price monitoring and Telegram long-polling without blocking the main thread.
- **Security & Performance:** Implemented custom API key authentication (with SHA-256 hashing), global rate limiting, and IMemoryCache to drastically reduce redundant external calls and prevent rate-limit bans.
- **Resilience & Data:** Engineered a resilient scraping mechanism with HTML parsing fallbacks (HtmlAgilityPack) for unreliable external APIs, persisting user sessions and alerts via Entity Framework Core and SQLite.
- **DevOps & Deployment:** Containerized the application using Docker and deployed it to an Oracle Cloud instance, reducing deployment setup from 2+ hours to under 5 minutes and ensuring reliable 24/7 production uptime.

Cripto Tracker | React / TypeScript / Tailwind 2025

- Developed a responsive Single Page Application (SPA) using **React**, **TypeScript**, and **Vite** to track real-time cryptocurrency prices via the CoinGecko API.
- Implemented a serverless backend using **Firebase** for secure user authentication (Google & Email/Password) and **Firestore** database to manage user data.
- Engineered a custom price alert system allowing users to set triggers (above/below targets) with dynamic fiat currency conversion (USD, BRL, EUR).
- Designed a modern, dark-mode UI using **Tailwind CSS**, featuring custom components like drawers, modals, and interactive data tables.

Professional Measurement Tool | Unreal Engine C++ (Fab / Unreal Marketplace, 2026) 2026

Author of a C++ editor plugin that measures distances and enclosed polygon areas in-editor using USplineComponent, with real-time visual feedback and a dedicated Editor Mode.

- Decoupled architecture using UInterface and multicast delegates, separating actor, widget, snap, and label logic into independent components.
 - Three snap modes (Off / Ground / Surface) with six-axis cardinal traces for precise point placement on complex geometry.
 - Custom Editor Mode (FEdMode) with Slate toolkit panel for one-click spawning, selection, and property editing.
- [Project link on FAB: <https://www.fab.com/listings/5c2a20d1-1537-4ca3-9255-405d38b79b05>]

Runtime Skinned Mesh Exporter (Fab / Unreal Marketplace, 2025) 2025

Author of a C++/Blueprint plugin that exports skinned Skeletal & Static Meshes at runtime to an OBJ + MTL + PNG package.

- Fully asynchronous pipeline: GPU→CPU transfer through FRHIGPUTextureReadback and PNG compression on the thread-pool, ensuring hitch-free operation.
 - Compatible with UE 4.27 → 5.6 via version-conditional code;
 - Managed the entire product lifecycle: ideation, development, documentation and multiproject testing
- [Project link on FAB: [Runtime Skeletal \(Skinned\) / Static Mesh Exporter | Fab](#)]

Real-Time Object Obstruction Detector | Unreal Engine C++ 2025

Built a high-performance C++ computer vision algorithm to compute real-time occlusion percentages between virtual objects in Unreal Engine.

- Replaced legacy offline analysis with a custom runtime solution, reducing validation time from around 2 hours to seconds.
- Maintains precise spatial measurements while preserving engine performance.

LANGUAGES

- **Languages:** English (C2), Portuguese (Native)

EDUCATION

